DETAILED COMPETITION PAPER OUTLINE

Curiosity Cup 2026: North Carolina EV Infrastructure Gap Analysis

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PAPER METADATA

Target Length: 4-5 pages main text + 1-2 page appendix
Category: Business & Economy
Primary Focus: Explosive EV adoption growth vs. lagging infrastructure deployment
Analytical Framework: 4-Phase (Descriptive → Diagnostic → Predictive → Prescriptive)
Software: SAS Studio, SAS Viya, Python embedded in SAS

SECTION 1: INTRODUCTION

Length: ½ - ¾ page
Tone: Accessible, problem-framing, sets stakes

Content Structure:

Opening Hook (2-3 sentences):

- North Carolina's electric vehicle adoption has exploded: 1,727% growth over 82 months
- This rapid transition creates a critical infrastructure challenge
- The question isn't whether EVs will dominate—it's whether charging infrastructure can keep pace

Problem Context (1 paragraph):

- Most EV charging occurs at home (80-90%), BUT public infrastructure remains system-critical for:
  - Long-distance travel and interstate corridors
  - Renters and multi-unit dwelling residents without home charging
  - Rural areas and underserved communities
  - Consumer confidence and range anxiety reduction

The Alignment Challenge (1 paragraph):

- EV adoption and infrastructure deployment are independent processes

- Misalignment creates risks:
  - Capacity strain (too many EVs, too few chargers)
  - Geographic inequality (urban abundance, rural scarcity)
  - Market inefficiency (overbuilt stations with low utilization)
  - Consumer barriers (range anxiety, charging anxiety)

Paper Preview (2-3 sentences):

- This study applies advanced analytics to quantify the gap between EV demand and charging supply
- Using county-level data across 82 months and 355 charging stations
- We identify where infrastructure lags, forecast future gaps, and provide evidence-based signals for investment prioritization

What NOT to include:

- ❌ NEVI funding details
- ❌ Policy recommendations (save for Impact section)
- ❌ Technical jargon or methodology details

Source Material:

- evpapersanyeretal.pdf: Intro framing structure
- teamdatamindcuriositycup2025.pdf: Tone and accessibility model

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SECTION 2: PROBLEM STATEMENT

Length: ½ page
Tone: Clear, specific, measurable

Content Structure:

Core Problem (1 short paragraph):

- North Carolina lacks a forward-looking, county-level understanding of when and where electric vehicle growth begins to outpace public charging capacity
- Current planning relies on reactive deployment rather than predictive analytics

- Infrastructure investment decisions are capital-intensive (>$400K per station) and path-dependent

Why This Matters - Three Angles (3 short paragraphs):

Economic Efficiency:

- Underbuilding constrains EV adoption and economic opportunity
- Overbuilding wastes capital and creates stranded assets
- Optimal deployment requires matching capacity to projected demand

Geographic Equity:

- Infrastructure concentrated in wealthy urban centers
- Rural and disadvantaged communities risk being left behind
- Two-tier transportation system emerging: urban EVs, rural ICE

System Readiness:

- EV growth alone is not the challenge—alignment determines readiness
- Gap between demand and supply creates friction in market transition
- Data-driven planning can accelerate electrification while controlling costs

Research Questions (2-3 bullets):

- Where do current infrastructure gaps exist across North Carolina counties?
- How will these gaps evolve if EV growth continues at current pace?
- Which counties face the most severe capacity constraints?

Source Material:

- evpapersanyeretal.pdf: Problem framing
- ncevregistrationseda.pdf: Evidence of explosive growth
- enhancedchargingunitseda.pdf: Infrastructure deployment patterns

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SECTION 3: DATA

Length: ½ page
Tone: Factual, transparent about sources and limitations

Content Structure:

Dataset 1: EV Registrations (Demand Side)

- **Source:** North Carolina Department of Transportation (NCDOT) vehicle registration database
- **Structure:** 8,200 observations (100 counties × 82 months)
- **Coverage:** September 2018 - June 2025 (~6.8 years)
- **Focus:** Battery Electric Vehicles (BEVs) only—plug-in hybrids excluded
- **Role:** Quantifies demand growth trajectory and geographic distribution

Dataset 2: Public Charging Infrastructure (Supply Side)

- **Source:** Alternative Fuels Data Center (AFDC), U.S. Department of Energy
- **Structure:** 1,725 charging unit records across 355 stations
- **Coverage:** 135 cities statewide, deployed 2011-2025
- **Detail:** Connector-level data with power output (kW) by type (NACS, CCS, CHAdeMO)
- **Role:** Quantifies charging capacity and geographic infrastructure coverage

Data Integration Strategy:

- Demand and supply datasets analyzed independently first
- Geocoded charging stations to county level using lat/long coordinates
- Joined at county level to calculate gap metrics
- Snapshot comparison (June 2025 BEVs vs. current infrastructure)

Data Limitations (Brief, Honest):

- Charging station data represents snapshot, not reliable time series
- No utilization data (actual charging sessions) available
- June 2025 BEV data potentially incomplete
- Missing demographic covariates (income, population density)

Source Material:

- ncevregistrationseda.pdf: Dataset 1 description
- enhancedchargingunitseda.pdf: Dataset 2 description
- Both EDAs: Structure and coverage details

SECTION 4: DATA CLEANING & VALIDATION

Length: ½ page
Tone: Professional, demonstrates rigor without overwhelming detail

Content Structure:

EV Registration Data Preparation:

- Validated temporal completeness: All 100 counties present in all 82 months (perfectly balanced panel)
- Standardized county identifiers (FIPS codes for join accuracy)
- Verified zero missing values across Electric vehicle field
- Excluded PlugInHybrid data per analysis scope

Charging Infrastructure Data Preparation:

- **Geocoding Process:** Converted 355 station lat/long coordinates to county names using NC county boundaries
- Validated connector type classifications (NACS, CCS, CHAdeMO, J1772)
- Calculated total charging capacity by station (sum of connector_count × power_output_kW)
- Verified geographic coverage: 135 cities mapped to XX counties

Data Integration & Quality Controls:

- County-level aggregation: Summed stations, chargers, and kW capacity per county
- Join validation: Matched all 100 counties (some with zero infrastructure)
- Consistency checks: BEV counts ≥ 0, charging capacity ≥ 0
- Sanity flags: Identified XX counties with >0 BEVs but zero public charging

Outcome:

- Analysis-ready county-month panel for BEV registrations
- Analysis-ready county-level charging capacity snapshot
- Clean join keys enabling gap analysis

What NOT to include:

- ❌ Code snippets
- ❌ Detailed technical process
- ❌ Over-explanation of basic cleaning

Source Material:

- ncevregistrationseda.pdf: Data quality assessment section
- enhancedchargingunitseda.pdf: Data quality assessment section
- Execution plan: Geocoding methodology

SECTION 5: ANALYSIS ⭐ CORE SECTION

Length: 1½ - 2 pages (the heart of the paper)
Tone: Analytical rigor balanced with accessibility

5.1 DESCRIPTIVE ANALYSIS (½ page)

Purpose: Establish the magnitude of EV growth and current infrastructure state

Demand Story - Explosive Growth (1 paragraph + 1 key stat box):

- Statewide BEV registrations grew from 5,165 (Sept 2018) to 94,371 (June 2025)
- **1,727% total growth**, **53.8% compound annual growth rate (CAGR)**
- Acceleration visible: early phase (2018-2020) grew at 2.71% MoM, peak phase (2021-2022) at 3.89% MoM
- Recent deceleration to 2.62% MoM suggests market maturation

Geographic Concentration (1 paragraph):

- Extreme inequality: Top 10 counties hold 72% of all BEVs (Gini coefficient = 0.805)
- Wake County leads: 25,032 BEVs (26.5% of state total)
- Urban dominance: 11 urban counties account for 70% of BEVs despite being only 11% of counties
- Research Triangle (Wake + Durham + Orange): 35.2% of state's EVs

Supply Story - Infrastructure Snapshot (1 paragraph):

- Current infrastructure: 355 charging stations, 1,725 individual charging units
- Total capacity: XX,XXX kW statewide (calculate from EDA)
- Technology mix: NACS (Tesla) 57.5%, CCS 37%, CHAdeMO 11.4%
- High-power focus: 80.5% of units provide ≥150 kW (DC fast charging)

Initial Imbalance Evidence (1-2 sentences):

- Statewide average: XX BEVs per station, YY BEVs per charger
- But massive county-level variation (to be explored in Diagnostic)

Key Visualization (referenced, in appendix):

- Figure A1: BEV growth trajectory (time series line chart)
- Figure A2: County-level BEV distribution (choropleth map)
- Figure A3: Charging station geographic distribution (dot map)

Source Material:

- ncevregistrationseda.pdf: All demand-side stats
- enhancedchargingunitseda.pdf: All supply-side stats
- evpapersanyeretal.pdf: Descriptive analysis structure

5.2 DIAGNOSTIC ANALYSIS (½ - ¾ page)

Purpose: Identify WHERE infrastructure gaps exist and HOW SEVERE

Multi-Metric Gap Framework (intro paragraph):

- Simple counts can mislead (1 Tesla Supercharger hub ≠ 1 small urban station)

- We employ three gap metrics to capture different dimensions:

Metric 1: BEVs per Station (Accessibility)

- County-level calculation: BEV_count / Station_count
- Identifies counties with sparse station coverage
- Top 5 worst: [List counties with highest ratios]
- Best coverage: [List counties with lowest ratios]

Metric 2: BEVs per kW Capacity (Utilization Pressure) ⭐ SOPHISTICATED

- County-level calculation: BEV_count / Total_Charging_Capacity_kW
- Accounts for station size variation (Tesla hubs vs. small sites)
- More accurate measure of actual charging strain
- Top 5 worst: [List counties with highest BEVs/kW]

Metric 3: Technology Adequacy (Future-Readiness)

- % of county capacity that's CCS (industry standard) vs. CHAdeMO (obsolete)
- High-power availability: % of capacity ≥150 kW
- Counties at risk: High reliance on legacy CHAdeMO infrastructure

Geographic Equity Analysis (1 paragraph):

- **Urban counties (11 total):**
  - Average BEVs per station: XX
  - Average BEVs per kW: YY
- **Rural counties (89 total):**
  - Average BEVs per station: XX (much higher = worse)
  - Average BEVs per kW: YY (much higher = worse)
- **Equity gap ratio:** Rural BEVs/kW is ZZ times worse than urban

Zero-Infrastructure Counties (if applicable):

- XX counties have >0 BEVs but zero public charging stations

- Total BEVs in underserved counties: XXX vehicles
- These represent complete infrastructure gaps

County Classification (1 paragraph): Segment counties into gap severity tiers:

- **Well-Aligned:** BEVs/kW below statewide median, adequate technology
- **Emerging Strain:** BEVs/kW 1-2x statewide median, moderate gaps
- **High Strain:** BEVs/kW >2x statewide median, severe capacity constraints
- **No Public Access:** Zero charging infrastructure despite EV presence

Key Visualization (referenced, in appendix):

- Figure A4: County gap severity map (choropleth with 4-tier classification)
- Figure A5: Urban vs. Rural gap comparison (bar chart)

Source Material:

- Derived from joining ncevregistrationseda + enhancedchargingunitseda datasets
- evpapersanyeretal.pdf: Diagnostic analysis methodology
- Execution plan: Gap metric calculations

5.3 PREDICTIVE ANALYSIS (½ page)

Purpose: Forecast future gaps if current trends continue

BEV Growth Projection (1 paragraph):

- Applied time-series forecasting to county-level registration data
- Methodology: [ARIMA / Exponential Smoothing / method from ncevregistrationseda.pdf]
- Statewide forecast: XX,XXX BEVs by December 2028 (XX% growth from June 2025)
- County-level forecasts show continued concentration in urban areas

Infrastructure Growth Scenarios (1 paragraph):

- **Challenge:** Snapshot data limits infrastructure forecasting
- **Approach:** Model three scenarios based on recent deployment patterns:

Scenario 1: Status Quo (Infrastructure Frozen)

- Assume current 355 stations, XX,XXX kW capacity remains static
- Future gap: Forecasted BEVs / Current capacity
- Result: Catastrophic strain by 2028

Scenario 2: Linear Continuation

- Recent deployment: 88 stations added in 2025 (24.8% of total infrastructure)
- Assume: +75-90 stations/year through 2028
- Future gap: Moderate improvement but still widening

Scenario 3: Demand-Matched Growth

- Calculate infrastructure needed to maintain current BEVs/kW ratio
- Required: +XXX stations, +XX,XXX kW capacity by 2028
- Investment implication: $XX million (at ~$400K per station)

Projected Gap by County Tier (1 paragraph):

- High-growth counties (Wake, Mecklenburg): Gap widens under Scenario 1 & 2
- Emerging markets (Franklin, Harnett): First-mover advantage window closing
- Rural counties: Already-severe gaps become extreme

Uncertainty Acknowledgment:

- Forecasts include confidence intervals (±2σ widening over time)
- Assumptions: Current growth rates persist, no major policy shocks
- Sensitivity: If BEV adoption accelerates/decelerates by X%, gap changes by Y%

Key Visualization (referenced, in appendix):

- Figure A6: Forecasted BEV growth with scenarios (ribbon chart)
- Figure A7: Projected gap by county tier (scenario comparison bars)

Source Material:

- ncevregistrationseda.pdf: Forecasting methodology and results
- evpapersanyeretal.pdf: Predictive analysis structure
- Execution plan: Scenario assumptions

5.4 PRESCRIPTIVE ANALYSIS ⚠️ DECISION POINT

Options:

Option A: Include Brief Prescriptive (¼ page)

- High-level framework only (no detailed optimization)
- "Data-driven signals for prioritization" approach
- Example: Priority counties = High BEV growth + High current gap + Low current capacity

Option B: Move to "Future Study" Section

- If page budget is tight
- Acknowledge prescriptive optimization as logical next step
- Keep Analysis section focused on descriptive/diagnostic/predictive

Recommendation: Option A - Include brief prescriptive to complete 4-phase framework, but keep it conceptual rather than mathematical

If including - Content:

Investment Prioritization Framework (1 paragraph):

- Counties should be prioritized based on composite score:
  - Current gap severity (from Diagnostic)
  - Forecasted demand growth (from Predictive)
  - Equity considerations (rural/disadvantaged status)
- High-priority counties: [Top 5-10 based on composite score]

Intervention Strategies by County Type (1 paragraph):

- **High-demand urban:** Densification (add chargers to existing stations)

- **Emerging growth:** Proactive deployment ahead of demand curve
- **Rural/corridor:** Strategic coverage along highways, tourism routes
- **Zero-infrastructure:** Seed installations to enable market entry

NOT included (to stay within guardrails):

- ❌ Specific funding programs or budgets
- ❌ Location-allocation optimization models
- ❌ Detailed cost-benefit calculations
- ❌ Policy prescriptions beyond "evidence-based signals"

Source Material:

- evpapersanyeretal.pdf: Prescriptive analysis framework (high-level only)

SECTION 6: VISUALIZATION

Length: 2-3 sentences in main text (figures in appendix)
Tone: Brief reference, not descriptive

Content:

In Main Text: "Supporting visualizations are provided in Appendix A, including time-series growth trajectories (Figure A1), county-level gap severity maps (Figure A4), and scenario-based projections (Figure A6). These visual representations reinforce the analytical findings and provide accessible insights for non-technical stakeholders."

In Appendix A (1-2 pages):

Essential Figures (must include):

1. **Figure A1:** BEV growth trajectory Sept 2018 - June 2025 (line chart)
2. **Figure A4:** County gap severity map (choropleth, 4-tier classification)
3. **Figure A6:** Forecasted BEV growth with infrastructure scenarios (ribbon chart)

Optional Figures (if space permits): 4. **Figure A2:** County BEV distribution heat map 5. **Figure A3:** Charging station locations (dot map overlaid on counties) 6. **Figure A5:** Urban vs. Rural gap comparison (bar chart) 7.

Figure A7: Network provider market share (pie chart - Tesla dominance) 8. **Figure A8:** Technology mix by county (stacked bar - NACS vs CCS vs CHAdeMO)

Source Material:

- All three EDA reports contain visualization examples
- teamdatamindcuriositycup2025.pdf: Appendix structure model

SECTION 7: IMPACT & IMPLICATIONS

Length: ½ page
Tone: Forward-looking, policy-relevant, avoids specific recommendations

Content Structure:

Current State Insights (1 paragraph):

- Infrastructure alignment exists in [X counties / Y% of counties]
- Severe gaps identified in [A counties], particularly rural areas
- Urban counties benefit from concentrated deployment but face utilization pressure
- Technology transition underway: NACS dominance (57.5%) signals market shift

Emerging and Projected Gaps (1 paragraph):

- If current trends persist, gap widens in XX counties by 2028
- High-growth emerging markets (Franklin, Harnett, Pender) require proactive deployment
- Rural access gap threatens equitable transition
- Scenario analysis shows demand-matched infrastructure requires $XX million investment

Planning Relevance (1 paragraph):

- Evidence-based signals enable targeted, efficient investment
- County-level gap metrics support phased deployment strategies
- Capacity-weighted analysis improves upon simple station counts
- Equity considerations ensure no communities left behind

Stakeholder Value (3-4 bullets):

- **Policy Makers:** Data-driven prioritization framework for infrastructure funding
- **Transportation Planners:** County-level forecasts support long-range planning
- **Utility Companies:** Demand projections inform grid capacity planning
- **Network Operators:** Gap analysis identifies underserved markets
- **Consumers:** Infrastructure visibility reduces range anxiety

Methodological Contribution (1 sentence): This study demonstrates scalable analytics framework applicable to other states/regions

Source Material:

- evpapersanyeretal.pdf: Impact section structure
- teamdatamindcuriositycup2025.pdf: Policy implications tone

SECTION 8: SUGGESTIONS FOR FUTURE STUDY

Length: ¼ - ½ page
Tone: Identifies logical extensions without diminishing current work

Content Structure:

Data Enhancement Opportunities:

- **Utilization data integration:** Actual charging session counts and kWh dispensed would enable demand-supply matching beyond capacity metrics
- **Demographic overlay:** Income, population density, and disadvantaged community designations would strengthen equity analysis
- **Temporal infrastructure data:** Reliable historical station deployment would enable supply-side forecasting

Analytical Extensions:

- **Scenario sensitivity analysis:** Model alternative EV adoption rates (accelerated, decelerated) and infrastructure growth trajectories
- **Corridor-specific analysis:** Deep dive on interstate highway charging adequacy for long-distance travel

- **Cost-benefit optimization:** Prescriptive modeling with capital constraints and utilization targets

Policy Integration:

- **NEVI funding allocation:** Evaluate federal infrastructure program deployment against identified gaps
- **Equity frameworks:** Incorporate Justice40 criteria and disadvantaged community prioritization
- **Private sector coordination:** Analyze public vs. private station deployment patterns

Methodological Advancement:

- **Real-time monitoring dashboard:** Operationalize gap metrics for ongoing tracking
- **Machine learning applications:** Predict optimal station locations using broader feature sets
- **Cross-state comparison:** Extend framework to multi-state analysis for regional planning

Source Material:

- All three EDA reports: "Next Steps for Deeper Analysis" sections
- evpapersanyeretal.pdf: Future study suggestions
- teamdatamindcuriositycup2025.pdf: Future research framing

SECTION 9: CONCLUSION

Length: ¼ - ½ page
Tone: Confident, synthesizes key messages, circles back to intro

Content Structure:

Restate Core Finding (2-3 sentences):

- North Carolina's electric vehicle adoption has grown explosively—1,727% over 82 months—but public charging infrastructure has not kept pace
- Our analysis quantifies this gap at the county level, revealing where capacity constraints exist today and where they will emerge by 2028
- The challenge is not whether EVs will grow, but whether infrastructure deployment can maintain alignment with demand

Key Insights Recap (3-4 bullets, brief):

- **Magnitude:** 94,371 BEVs (June 2025) face uneven charging access across 100 counties
- **Geography:** Extreme urban-rural disparity—urban counties have XX times better access than rural counties
- **Technology:** Capacity-weighted metrics reveal Tesla's dominance (60.5% of connectors) and legacy CHAdeMO burden
- **Trajectory:** Without accelerated deployment, gap widens significantly by 2028

Methodological Value Statement (1-2 sentences):

- This study demonstrates that data-driven, multi-metric gap analysis provides a scalable foundation for infrastructure planning
- The 4-phase analytical framework (descriptive, diagnostic, predictive, scenario-based) offers replicable approach for other states and regions

Closing Message - Alignment Over Growth (2-3 sentences):

- EV growth alone does not determine electrification success—infrastructure alignment determines system readiness
- Policymakers, planners, and investors require evidence-based signals to deploy capital efficiently and equitably
- By quantifying current gaps and forecasting future needs, data analytics transforms reactive infrastructure planning into proactive, strategic investment

Final Thought (1 sentence): North Carolina's EV transition is well underway; ensuring infrastructure keeps pace will determine whether this transition is efficient, equitable, and accessible to all.

Source Material:

- evpapersanyeretal.pdf: Conclusion structure
- teamdatamindcuriositycup2025.pdf: Closing tone
- Introduction section: Circle back to opening themes

APPENDIX A: SUPPORTING VISUALIZATIONS

Length: 1-2 pages

Content: Figures referenced throughout main text

Minimum Required:

- Figure A1: BEV growth trajectory (time series)
- Figure A4: County gap severity map (choropleth)
- Figure A6: Forecasted scenarios (ribbon chart)

Additional if space permits:

- Figures A2, A3, A5, A7, A8 (as listed in Section 6)

CRITICAL ELEMENTS CHECKLIST

✅ Competition Requirements:

- [ ] 2-5 pages main text (target 4)
- [ ] Optional 2-page appendix (use for visualizations)
- [ ] All required sections present
- [ ] SAS software statement: "Analysis conducted using SAS Studio, SAS Viya, and Python embedded in SAS for geospatial processing"
- [ ] Accessible to non-technical judges
- [ ] Original work, properly cited

✅ Analytical Rigor:

- [ ] 4-phase framework clearly executed
- [ ] Multi-metric gap analysis (not just simple counts)
- [ ] County-level granularity throughout
- [ ] Forecasting with scenarios
- [ ] Equity dimension included

✅ Story Arc:

- [ ] Introduction frames problem
- [ ] Analysis proves the gap exists and quantifies it
- [ ] Impact shows why it matters
- [ ] Conclusion ties back to opening

✅ **Guardrails Respected:**

- [ ] No funding program details
- [ ] No specific policy recommendations
- [ ] No location-allocation optimization details
- [ ] Stay analytical, not prescriptive in policy terms

PAGE ALLOCATION SUMMARY

Section	Pages	Priority
Introduction	0.5-0.75	HIGH
Problem Statement	0.5	HIGH
Data	0.5	MEDIUM
Data Cleaning	0.5	MEDIUM
Analysis (5.1-5.4)	**1.5-2.0**	**CRITICAL**
Visualization (text only)	0.05	LOW
Impact & Implications	0.5	HIGH
Future Study	0.25-0.5	MEDIUM
Conclusion	0.25-0.5	HIGH
TOTAL MAIN TEXT	**4-5 pages**	
Appendix A (figures)	**1-2 pages**	MEDIUM

DECISION POINTS FOR REVIEW

1. Prescriptive Analysis:

- Include brief version (¼ page) in Section 5.4? **OR**
- Move to Future Study section to save space?

- **Recommendation:** Include brief version to complete 4-phase framework

2. Appendix Size:

- Minimal (3 essential figures in 1 page)? **OR**
- Comprehensive (8 figures in 2 pages)?
- **Recommendation:** Start with 3 essential, add more if analysis warrants

3. Technical Depth in Analysis Section:

- Keep methodology high-level for accessibility? **OR**
- Include more technical detail (equations, model specifics)?
- **Recommendation:** High-level in main text, technical details in appendix footnotes if needed

4. Equity Analysis Prominence:

- Brief mention in Diagnostic section? **OR**
- Dedicated subsection within Diagnostic?
- **Recommendation:** Dedicated paragraph in Diagnostic (important differentiator)

STATUS: Outline complete and ready for review
NEXT STEP: Create Updated Execution Plan with geocoding workflow and analysis sequence